are now speeding away, you would expect to measure a slower speed for the oncoming photons. Specifically, you would expect to find them approaching you at (670 million miles per hour - 100 million miles per hour =) 570 million miles per hour.

Mounting evidence from a variety of experiments dating back as far as the 1880s, as well as careful analysis and interpretation of Maxwell's electromagnetic theory of light, slowly convinced the scientific community that, in fact, this is not what you will see. Even though you are retreating, you will still measure the speed of the approaching photons as 670 million miles per hour, not a bit less. Although at first it sounds completely ridiculous, unlike what happens if one runs from an oncoming baseball, grenade, or avalanche, the speed of approaching photons is always 670 million miles per hour. The same is true if you run toward oncoming photons or chase after them—their speed of approach or recession is completely unchanged; they still appear to travel at 670 million miles per hour. Regardless of relative motion between the source of photons and the observer, the speed of light is always the same.

Technological limitations are such that the "experiments" with light, as described, cannot actually be carried out. However, comparable experiments can. For instance, in 1913 the Dutch physicist Willem de Sitter suggested that fast-moving binary stars (two stars that orbit one another) could be used to measure the effect of a moving source on the speed of light. Various experiments of this sort over the past eight decades have verified that the speed of light received from a moving star is the same as that from a stationary star—670 million miles per hour—to within the impressive accuracy of ever more refined measuring devices. Moreover, a wealth of other detailed experiments has been carried out during the past century—experiments that directly measure the speed of light in various circumstances, as well as test many of the implications arising from this characteristic of light, as discussed shortly—and all have confirmed the constancy of the speed of light.

If you find this property of light hard to swallow, you are not alone. At the turn of the century physicists went to great length to refute it. They couldn't. Einstein, to the contrary, embraced the constancy of the speed of light, for here was the answer to the paradox that had troubled